CURRICULUM VITAE

Ryo Akiba

Undergraduate researcher · Institute of Science Tokyo

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Education

Present **Institute of Science Tokyo**
Fourth-year undergraduate, School of Life Science and Technology
Undergraduate thesis research on RNA tertiary structure prediction at Sato Laboratory.

Research Interests

Structure prediction · Function prediction · Protein · RNA · AI for Science

Research & Professional Experience

Computational Drug Discovery Laboratory

Research Assistant Aug 2026 – Present

- Multi-objective optimization for designing structurally similar proteins with dissimilar sequences.
- Research on improving the chemical validity of ligand structures generated in protein–ligand complexes.

Preferred Networks, Inc.

Engineer Sep 2025 – Jul 2026

- Research on the automated construction and analysis of a benchmark for AI-assisted troubleshooting of biological experiments (under review).

Summer Intern Jul 2025 – Sep 2025

- Worked on AI-assisted troubleshooting of biological experiments.

Publication

Multi-objective optimization for designing structurally similar proteins with dissimilar sequences

Ryo Akiba, Yoshitaka Moriwaki, Ryuichiro Ishitani, and Naruki Yoshikawa.

Science and Technology of Advanced Materials: Methods, 6(1), 2611575 (2026).

doi:10.1080/27660400.2025.2611575

Selected Research Project

AI-assisted troubleshooting of biological experiments

Developed an LLM-based agent workflow that identifies errors in biological experiment protocols by generating hypotheses and validating them against external knowledge.

Technical blog, Preferred Networks

Ryo Akiba

Presentations

2026

Oct 31, 2026 · Oral presentation · Upcoming
Improving RNA template search and residue alignment accuracy using predicted structures
10th Tokyo Bioinformatics Meeting · Keio University, Yagami Campus

Oct 31, 2026 · Poster presentation · Upcoming
Template-assisted machine learning for RNA tertiary structure prediction without additional training
10th Tokyo Bioinformatics Meeting · Keio University, Yagami Campus

Oct 2 – 4, 2026 · Poster presentation
Investigating the use of template information in deep learning-based RNA 3D structure prediction
18th Annual Meeting of Young Researchers in Bioinformatics · Shonan Village Center, Kanagawa

Jun 19, 2026 · Oral presentation
Multi-objective optimization for designing structurally similar proteins with dissimilar sequences
AI4X-Accelerate 2026 · Raffles City Convention Centre, Singapore

2025

Dec 10, 2025 · Oral presentation
Multi-objective optimization for designing structurally similar proteins with dissimilar sequences
1st Biomolecular Design Workshop · Institute of Science Tokyo, Yushima Campus

Oct 30, 2025 · Oral presentation · Oral Presentation Award
Multi-Objective Optimization for Designing Structurally Similar Proteins with Diverse Sequence
CBI Annual Meeting 2025 · Tower Hall Funabori, Tokyo

Activities

2024 **iGEM TokyoTech 2024 · Sub-leader**
RNA pesticide delivery using engineered membrane vesicles.

2023 **iGEM TokyoTech 2023 · Dry Lab member**
Low-cost, high-throughput assays for dengue surveillance.

Honors & Awards

2025 Oral Presentation Award, CBI Annual Meeting 2025

2025 Judges' Award, Lean Launchpad National Competition 2025 Kyoto (iGEM TokyoTech 2024 team)

2024 iGEM Gold Medal (TokyoTech team)

2023 iGEM Silver Medal (TokyoTech team)